

Encoding MP3s

Lame is the most popular encoder used, which can be found at [audio/lame](http://audio.lame.org) port. To convert a ripped WAV file, use the following command:

```
• # lame -h -b 128 \
• --tt "Foo Song Title" \
• --ta "FooBar Artist" \
• --tl "FooBar Album" \
• --ty "2001" \
• --tc "Ripped and encoded by Vik" \
• --tg "Genre" \
• audio01.wav audio01.mp3
```

This will convert the WAV file to an MP3 file.

Decoding MP3s

Conversely, to burn an audio CD from MP3, you must first convert it into a non-compressed WAV format. XMMS and mpg123 can both be used to convert mp3 into a WAV file. To use mpg123, specify the name of the PCM file:

```
• #mpg123 -s audio007.mp3 > audio007.pcm
```

To use XMMS, follow these steps:

1. Launch XMMS.
2. Bring up the XMMS menu by right-clicking.
3. Under **Options** select **Preferences**.
4. Change the output plugin to **Disk Writer Plugin**.
5. Press **Configure**.
6. Select the directory where you want the uncompressed files to be written.
7. Load the MP3 file into XMMS.
8. Adjust the settings such as the volume (100%) and EQ (turned off).
9. Press **Play**.
10. When done, change the output plugin to what it was before.

WAV, as well as PCM audio formats, can be used with `cdrecord` to create audio CDs.

Image scanners

Access to image scanners is provided by the Scanner Access Now Easy (SANE) API. FreeBSD supports SCSI and USB scanners. Before performing any configuration, make sure that SANE supports the scanner. You can also check the list of supported devices before proceeding with the configuration process.